

Analyses: Classification with KNN

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Description

k-nearest neighbor analyses of the feature vectors per character sequence (loaded from `~/results/features_combined.csv`). The results are stored in `~/results/classifier/`.

Load Packages

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(dplyr)
library(class)
library(gmodels)
library(caret)
```

Load Data

Load data table with estimated feature values per sequence.

```
# load feature value estimations
estimations.df <- read.csv("~/Github/UpperPalCombinatorics/results/features_combined.csv")
head(estimations.df)
```

```
##      subcorpus      sequence num.char huni.chars hrate.chars  ttr.chars
## 1 Aurignacien ||| ///// _ V      12  1.8962406   1.5295563  0.41666667
## 2 Aurignacien      vvvvv         5  0.0000000   0.5711903  0.20000000
## 3 Aurignacien      vvvv vvvv      9  0.5032583   0.9060047  0.22222222
## 4 Aurignacien vvvvvvvvvvvvvvv      14  0.0000000   0.6379550  0.07142857
## 5 Aurignacien      |||           3  0.0000000   0.4308271  0.33333333
## 6 Aurignacien      |||           4  0.0000000   0.5070802  0.25000000
##      rm.chars
## 1 0.6363636
## 2 1.0000000
## 3 0.7500000
## 4 1.0000000
## 5 1.0000000
## 6 1.0000000
```

Select subcorpora for binary classification.

```
# choose subcorpora to compare
unique(estimations.df$subcorpus)
```

```
## [1] "Aurignacien"          "Cuneiform (Uruk V)"    "Cuneiform (Uruk IV)"
## [4] "Cuneiform (Uruk III)" "TeDDi"
```

```
subcorpus1 <- "Cuneiform (Uruk III)"
subcorpus2 <- "Aurignacien"
selected <- c(subcorpus1, subcorpus2)
estimations.df <- estimations.df[estimations.df$subcorpus %in% selected, ]
```

Select relevant columns of the data frame, i.e. the measures to be included in classification and the “subcorpus” column.

```
estimations.subset <- estimations.df[c("subcorpus",
                                       "huni.chars",
                                       "hrate.chars",
                                       "ttr.chars",
                                       "rm.chars"
                                       )]
```

Remove NAs (whole row)

```
estimations.subset <- na.omit(estimations.subset)
```

Center and scale the data

```
# first column is subcorpus column, the others are the features
# be aware that this needs to be changed in case of other table format
estimations.scaled <- cbind(estimations.subset["subcorpus"],
                           scale(estimations.subset[, c("huni.chars", "hrate.chars",
                                                         "ttr.chars", "rm.chars")]))
```

Create Training and Test Sets

```
# Generating seed
set.seed(1234)
# Randomly generating our training and test samples with a respective ratio of 2/3 and 1/3
datasample <- sample(2, nrow(estimations.scaled), replace = TRUE, prob = c(0.67, 0.33))

# Generate training set (select rows randomly by using datasample vector)
train <- estimations.scaled[datasample == 1, 1:ncol(estimations.scaled)]
# Generate test set
test <- estimations.scaled[datasample == 2, 1:ncol(estimations.scaled)]
```

Initialize data frame

```
knn.results <- data.frame(k = numeric(0), accuracy = numeric(0), precision = numeric(0),
                          recall = numeric(0), f1 = numeric(0))
```

Building knn classifier

```
# choose maximum number of neighbors n
n = 10
# run a loop over different numbers of neighbors up to n
for (k in 1:n){
  # knn estimation of labels
  predictions.knn <- knn(train = train[, c("huni.chars",
                                           "hrate.chars",
                                           "ttr.chars",
                                           "rm.chars")],
                        test = test[, c("huni.chars",
                                         "hrate.chars",
                                         "ttr.chars",
                                         "rm.chars")],
                        cl = train[, "subcorpus"], k = k)

  # model evaluation
  # creating a dataframe from known (true) test labels
  test.labels <- test[, "subcorpus"]
  # combining predicted and known classes
  class.comparison <- data.frame(predictions.knn, test.labels)
  # giving appropriate column names
  names(class.comparison) <- c("predicted", "observed")
  # inspecting our results table
  head(class.comparison)
  # get confusion matrix
  cm <- confusionMatrix(data = class.comparison$predicted,
                        reference = as.factor(class.comparison$observed))

  print(cm)
  # get precision, recall, and f1 from the output list of confusionMatrix()
  accuracy <- cm$overall['Accuracy']
  f1 <- cm[["byClass"]][["F1"]]
  recall <- cm[["byClass"]][["Recall"]]
  precision <- cm[["byClass"]][["Precision"]]

  # prepare data frame with results
  local.results <- data.frame(k, accuracy, precision, recall, f1, row.names = NULL)
  local.results.rounded <- round(local.results, 2)
  # print(local.results.rounded)
  knn.results <- rbind(knn.results, local.results.rounded)
}
```

```
## Confusion Matrix and Statistics
##
##               Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##   Aurignacien              49                  13
##   Cuneiform (Uruk III)      11                  277
##
##               Accuracy : 0.9314
##               95% CI : (0.8997, 0.9556)
##               No Information Rate : 0.8286
```

```

##      P-Value [Acc > NIR] : 1.239e-08
##
##      Kappa : 0.7618
##
##      McNemar's Test P-Value : 0.8383
##
##      Sensitivity : 0.8167
##      Specificity : 0.9552
##      Pos Pred Value : 0.7903
##      Neg Pred Value : 0.9618
##      Prevalence : 0.1714
##      Detection Rate : 0.1400
##      Detection Prevalence : 0.1771
##      Balanced Accuracy : 0.8859
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              46              14
##  Cuneiform (Uruk III)      14             276
##
##      Accuracy : 0.92
##      95% CI : (0.8865, 0.9462)
##      No Information Rate : 0.8286
##      P-Value [Acc > NIR] : 5.655e-07
##
##      Kappa : 0.7184
##
##      McNemar's Test P-Value : 1
##
##      Sensitivity : 0.7667
##      Specificity : 0.9517
##      Pos Pred Value : 0.7667
##      Neg Pred Value : 0.9517
##      Prevalence : 0.1714
##      Detection Rate : 0.1314
##      Detection Prevalence : 0.1714
##      Balanced Accuracy : 0.8592
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              47              13
##  Cuneiform (Uruk III)      13             277
##
##      Accuracy : 0.9257
##      95% CI : (0.893, 0.9509)
##      No Information Rate : 0.8286

```

```

##      P-Value [Acc > NIR] : 9.068e-08
##
##      Kappa : 0.7385
##
##      McNemar's Test P-Value : 1
##
##      Sensitivity : 0.7833
##      Specificity : 0.9552
##      Pos Pred Value : 0.7833
##      Neg Pred Value : 0.9552
##      Prevalence : 0.1714
##      Detection Rate : 0.1343
##      Detection Prevalence : 0.1714
##      Balanced Accuracy : 0.8693
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              47              13
##  Cuneiform (Uruk III)      13             277
##
##      Accuracy : 0.9257
##      95% CI : (0.893, 0.9509)
##      No Information Rate : 0.8286
##      P-Value [Acc > NIR] : 9.068e-08
##
##      Kappa : 0.7385
##
##      McNemar's Test P-Value : 1
##
##      Sensitivity : 0.7833
##      Specificity : 0.9552
##      Pos Pred Value : 0.7833
##      Neg Pred Value : 0.9552
##      Prevalence : 0.1714
##      Detection Rate : 0.1343
##      Detection Prevalence : 0.1714
##      Balanced Accuracy : 0.8693
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              46              11
##  Cuneiform (Uruk III)      14             279
##
##      Accuracy : 0.9286
##      95% CI : (0.8964, 0.9532)
##      No Information Rate : 0.8286

```

```

##      P-Value [Acc > NIR] : 3.422e-08
##
##      Kappa : 0.7435
##
##      McNemar's Test P-Value : 0.6892
##
##      Sensitivity : 0.7667
##      Specificity : 0.9621
##      Pos Pred Value : 0.8070
##      Neg Pred Value : 0.9522
##      Prevalence : 0.1714
##      Detection Rate : 0.1314
##      Detection Prevalence : 0.1629
##      Balanced Accuracy : 0.8644
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              45              11
##  Cuneiform (Uruk III)      15             279
##
##      Accuracy : 0.9257
##      95% CI : (0.893, 0.9509)
##      No Information Rate : 0.8286
##      P-Value [Acc > NIR] : 9.068e-08
##
##      Kappa : 0.7314
##
##      McNemar's Test P-Value : 0.5563
##
##      Sensitivity : 0.7500
##      Specificity : 0.9621
##      Pos Pred Value : 0.8036
##      Neg Pred Value : 0.9490
##      Prevalence : 0.1714
##      Detection Rate : 0.1286
##      Detection Prevalence : 0.1600
##      Balanced Accuracy : 0.8560
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              44              12
##  Cuneiform (Uruk III)      16             278
##
##      Accuracy : 0.92
##      95% CI : (0.8865, 0.9462)
##      No Information Rate : 0.8286

```

```

##      P-Value [Acc > NIR] : 5.655e-07
##
##      Kappa : 0.7107
##
##      McNemar's Test P-Value : 0.5708
##
##      Sensitivity : 0.7333
##      Specificity : 0.9586
##      Pos Pred Value : 0.7857
##      Neg Pred Value : 0.9456
##      Prevalence : 0.1714
##      Detection Rate : 0.1257
##      Detection Prevalence : 0.1600
##      Balanced Accuracy : 0.8460
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              45              11
##  Cuneiform (Uruk III)      15             279
##
##      Accuracy : 0.9257
##      95% CI : (0.893, 0.9509)
##      No Information Rate : 0.8286
##      P-Value [Acc > NIR] : 9.068e-08
##
##      Kappa : 0.7314
##
##      McNemar's Test P-Value : 0.5563
##
##      Sensitivity : 0.7500
##      Specificity : 0.9621
##      Pos Pred Value : 0.8036
##      Neg Pred Value : 0.9490
##      Prevalence : 0.1714
##      Detection Rate : 0.1286
##      Detection Prevalence : 0.1600
##      Balanced Accuracy : 0.8560
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              47              10
##  Cuneiform (Uruk III)      13             280
##
##      Accuracy : 0.9343
##      95% CI : (0.903, 0.9579)
##      No Information Rate : 0.8286

```

```

##      P-Value [Acc > NIR] : 4.294e-09
##
##      Kappa : 0.764
##
##      McNemar's Test P-Value : 0.6767
##
##      Sensitivity : 0.7833
##      Specificity : 0.9655
##      Pos Pred Value : 0.8246
##      Neg Pred Value : 0.9556
##      Prevalence : 0.1714
##      Detection Rate : 0.1343
##      Detection Prevalence : 0.1629
##      Balanced Accuracy : 0.8744
##
##      'Positive' Class : Aurignacien
##
## Confusion Matrix and Statistics
##
##              Reference
## Prediction      Aurignacien Cuneiform (Uruk III)
##  Aurignacien              45              10
##  Cuneiform (Uruk III)      15             280
##
##      Accuracy : 0.9286
##      95% CI : (0.8964, 0.9532)
##      No Information Rate : 0.8286
##      P-Value [Acc > NIR] : 3.422e-08
##
##      Kappa : 0.74
##
##      McNemar's Test P-Value : 0.4237
##
##      Sensitivity : 0.7500
##      Specificity : 0.9655
##      Pos Pred Value : 0.8182
##      Neg Pred Value : 0.9492
##      Prevalence : 0.1714
##      Detection Rate : 0.1286
##      Detection Prevalence : 0.1571
##      Balanced Accuracy : 0.8578
##
##      'Positive' Class : Aurignacien
##

```

```
print(knn.results)
```

```

##      k accuracy precision recall  f1
## 1      1      0.93      0.79      0.82 0.80
## 2      2      0.92      0.77      0.77 0.77
## 3      3      0.93      0.78      0.78 0.78
## 4      4      0.93      0.78      0.78 0.78
## 5      5      0.93      0.81      0.77 0.79
## 6      6      0.93      0.80      0.75 0.78
## 7      7      0.92      0.79      0.73 0.76

```

```
## 8 8 0.93 0.80 0.75 0.78
## 9 9 0.93 0.82 0.78 0.80
## 10 10 0.93 0.82 0.75 0.78
```

Write to file. Note that the file names have to be changed manually here.

```
write.csv(knn.results, file = paste(c("~/Github/UpperPalCombinatorics/results/classifier/KNN/results_KNN",
                                     subcorpus1, "_",
                                     subcorpus2, "_",
                                     ".csv"
                                     ), collapse = ""),
          row.names = F)
```